Lab 2: Waves in Homogeneous Media

ENEL 476: Electromagnetic Waves and Applications Prof. Janitz - Winter 2026

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1 Introduction

1.1 Uniform Plane Waves

Uniform plane waves (UPW) are a special case of electromagnetic radiation. UPWs are commonly used to model the propagation of electromagnetic waves in regions far away from sources. In this lab, we will investigate UPWs propagation in a homogeneous media.

A UPW propagating in the +z-direction is given by:

$$E(z,t) = E_0 e^{-\alpha z} \cos(\omega t - \beta z) \tag{1}$$

where E_0 is the amplitude in V/m, α is the attenuation coefficient in Np/m, β is the phase constant in rad/m, and ω is the angular frequency in rad/s. The UPW in phasor form is given by:

$$E(z) = E_0 e^{-(\alpha + j\beta)z} = E_0 e^{-\gamma z} \tag{2}$$

where $\gamma = \alpha + j\beta$ is the complex propagation constant.

The H-field can be calculated using the material instrinic impedance, η:

$$H(z,t) = \frac{E_0}{|\eta|} e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta) \tag{3}$$

or in phasor form:

$$H(z) = \frac{E_0}{|\eta|} e^{-(\alpha + j\beta)z - j\theta_\eta} \tag{4}$$

The following equations can be used to calculate the attenuation coefficient, phase constant, and material intrinsic impedance for an arbitrary medium:

$$\alpha = \omega \sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1 + \left[\frac{\sigma}{\omega\varepsilon}\right]^2} - 1\right)} \tag{5}$$

$$\beta = \omega \sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1 + \left[\frac{\sigma}{\omega\varepsilon}\right]^2} + 1\right)} \tag{6}$$

$$|\eta| = \sqrt{\frac{\mu}{\varepsilon}} \frac{1}{\left[1 + (\sigma/\omega\varepsilon)^2\right]^{\frac{1}{4}}} \tag{7}$$

$$\tan(2\theta_\eta) = \frac{\sigma}{\omega\varepsilon}, \text{ for } 0 \le \theta \le \pi/4 \tag{8}$$

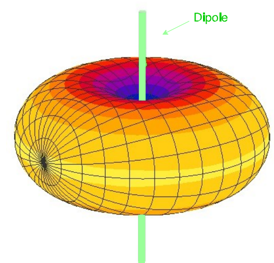

Figure 1: Far-field radiation pattern for a dipole antenna.

1.2 Half-Wavelength Dipole Antenna

Uniform plane waves are useful for modeling fields far away from sources, but fields must be generated from a source such as an antenna. The half wavelength dipole is the one of the simplest antennas to design and fabricate. The dipole is constructed of two quarter wavelength wires to make an antenna that is half a wavelength in length. At distances far from the antenna, the fields can be modelled as follows:

$$E(r, \phi, \theta) = j60I_0 \frac{e^{-j\beta r}}{r} \frac{\cos\left(0.5\pi \cos(\theta)\right)}{\sin(\theta)} \hat{a}_\theta \tag{9}$$

where I_0 is the peak current on the wire and r, ϕ, and θ are the spherical coordinates as defined in the course formula sheet. Fig. 1 shows the far field radiation pattern for a half-wavelength dipole.

1.3 Deliverables

The following are the deliverables for this lab:

1. Pre-Lab: Complete the pre-lab questions before the lab period. The prelab should be completed individually. The pre-lab will marked for completion and should be corrected before proceeding with the lab.

2. Lab: Submit a lab report in the Jupyter Notebook (.ipynb) on D2L. One report per group is required. The report is due 11:59 pm one week from the lab date. All work can be entered into the Notebook and uploaded.

A template is provided on D2L. Lab report expectations are as follows:

- Answer all questions and report all values in bold. (** value ** in Jupyter markdown.)

- Introduction, procedure, and conclusion are not required.

- Answer all questions in complete sentences, show work for any calculations, and provide tables and graphs with proper units, labels, captions as necessary. Equations can be entered in Jupyter markdown cells in two ways
  1. in-line: `$\eta = 5 \pi + 2$` results in $\eta = 5\pi + 2$, in-line with the text
  2. separate line: `$$\eta = 5 \pi + 2$$` results in
  $$\eta = 5\pi + 2$$
  on a separate text line.

Jupyter Notebook can be accessed through https://ucalgary.syzygy.ca using your UCID login information, or through the standard Anaconda Python installation. Note for Anaconda installation you may need to install the ffmpeg package for animations to work.

2 Pre-Lab

2.1 Uniform Plane Waves

Consider the geometry in Fig. 2 to answer the following questions.

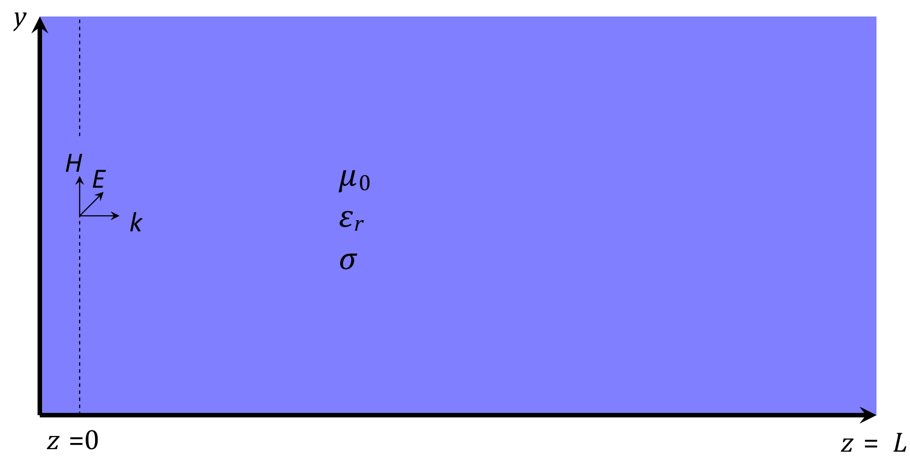

Figure 2: Uniform plane wave geometry.

1. Complete Table 1 by calculating α, β, λ and η for the given scenarios.

2.2 Half-Wavelength Dipole Antenna

1. Calculate the length of the dipole arms at 3 GHz.
2. Calculate the formula for the E_x, E_y, and E_z components of the electric field from a dipole.

Scenario	α	β	λ	η
Free space at 3 GHz				
Free space at 1.5 GHz				
$\varepsilon_r = 4$, $\sigma = 0.1$ S/m at 3 GHz				

Table 1: Uniform plane wave parameters for homogeneous media.

3 Lab

3.1 Uniform Plane Wave Visualization

The behavior of UPWs in homogeneous media will be visualized in Python.
Both the time-varying form of the E- and H-fields will be visualized in 3D. On D2L, the Jupyter Notebook template provides an example to help plot these functions. These scripts include functions to generate quiver plots as well as to combine plots into a .gif to make an animation.

- Frequency = 3 GHz
- $0 \leq z \leq 30$ cm
- $0 \leq t \leq 1$ ns
- Free space medium
- Wave traveling in the +z-direction with E-field in +x-direction.
- Initial field magnitude of 1 V/m

Using the above parameters, unless otherwise mentioned, write a script to do the following in free space:

1. Plot the electric and magnetic fields as a function of z at $t = 0$ in 3D.
2. Plot the electric and magnetic fields as a function of time at $z = 0$ in 2D.

3. Plot the magnitude of the electric and magnetic field phasors in 3D.
4. Generate an animation of the fields for $0 \leq t \leq 1$ ns in 3D.

Using this code, generate E- and H-field animations and phasor magnitude plots for the following scenarios:

5. Wave traveling in +z-direction in free space at 3 GHz.

6. Wave traveling in +z-direction in free space at 1.5 GHz.

7. Wave traveling in +z-direction in a homogeneous medium with $\varepsilon_r = 4$, $\mu_r = 1$, and $\sigma = 0.1$ S/m at 3 GHz.

8. Wave traveling in the -z-direction and E-field in the +x-direction in free space.

9. The sum of a forward-traveling wave with magnitude 1 and phase 0, and a backward-traveling wave with magnitude 0.5 and phase $\pi/4$ where both are traveling in free space and both have E-field oriented in the +xdirection.

Using these plots as a guide, answer the following questions:

10. What impact does increasing the permittivity have on the wavelength?

11. How does decreasing the frequency impact the wavelength?

12. What impact does increasing the permittivity have on the magnetic field strength?

13. How far does the wave travel before it undergoes $1Np$ of attenuation when $\varepsilon_r = 4$ and $\sigma = 0.1$ S/m.

14. Describe the behavior of the wave resulting from the sum of forward and backward traveling waves over time?

3.2 Half-Wavelength Dipole Antenna

The fields generated by a dipole antenna will be visualized in 3D to investigate more realistic wave propagation and demonstrate the practical creation of UPW far from the source.

First, write a script for calculating the x, y, and z components of the electric fields in Cartesian coordinates based on your pre-lab results.eq.

Next, plot the magnitude of the time-domain E-field (all components) in xy and xz planes in the region near the antenna ($0.1m \leq x \leq 2m$, $-1m \leq y \leq 1m$, $-1m \leq z \leq 1m$) at $t = 0$.

Some helpful programming tips:

- *plt.imshow*() in Python's matplotlib package can be used to plot 2D data.

- *np.meshgrid*() in Python's NumPy package can be used to generate matrices with coordinates *X,Y* from arrays of points *x,y*.
- *plt.colorbar*() can be used to add a colorbar with units to an *imshow*() plot.

Using these plots and code similar to part 3.1 generate animations of the fields in the xy and xz planes.

Generate a plot further from the antenna (e.g. $2m \le x \le 4m$). Finally, generate an animation of the fields very far away from the dipole in the xy plane (e.g. $100m \le x \le 102m$, $-1m \le y \le 1m$).

Answer the following considering your animations:

15. How does the E-field strength vary in the ϕ direction in the xy-plane?

16. How does the E-field strength vary as the fields move farther from the antenna?

17. Relative to the alignment of the dipole, in which direction is the E-field the weakest?

18. Compare the behavior of the wave generated by the dipole antenna to a UPW at: 1) $0 \le z \le 2$ m, 2) 2 m $\le z \le$ 4 m, and 3) 100 m $\le z \le$ 102 m.

4 Conclusion

Uniform plane waves are a simple model of radiating electromagnetic waves that are useful in a wide range of engineering problems. Visualizing the time-varying waves in 3D provides greater insight into their behavior. UPW can accurately model more realistic sources such as dipole antennas far away from the source.